

Lab-06: Implementation of a Composite Geometric Shape (Ice Cream Cone) using Line and Circle Drawing Algorithms.

Code:

```
main.cpp X main.cpp X
1  #include<windows.h>
2  #include <GL/glut.h>
3  #include <iostream>
4  #include <cmath>
5  using namespace std;
6
7  // DDA Line
8  void line(float xs, float xe, float ys, float ye){
9      float dx = xe - xs;
10     float dy = ye - ys;
11
12     float m = dy / dx;
13
14     if (abs(m) < 1)
15     {
16         if (xs > xe)
17         {
18             swap(xs, xe);
19             swap(ys, ye);
20         }
21
22         float y = ys;
23
24         for (float x = xs; x <= xe; x += 1)
25         {
26             glVertex2i(round(x), round(y));
27             y = y + m;
28         }
29     }
30     else
31     {
32         if (ys > ye)
33         {
34             swap(xs, xe);
35             swap(ys, ye);
36         }
37
38         float x = xs;
```

```
main.cpp X main.cpp X
37
38     float x = xs;
39
40     for (float y = ys; y <= ye; y += 1)
41     {
42         glVertex2i(round(x), round(y));
43         x = x + (1/m);
44     }
45 }
46
47 // Midpoint Circle
48 void circle(float r, float xc, float yc){
49     float x = 0;
50     float y = r;
51     float p = 1 - r;
52
53     while (x <= y)
54     {
55         // 8-way symmetry
56         glVertex2i(xc + x, yc + y);
57         glVertex2i(xc + y, yc + x);
58         glVertex2i(xc + y, yc - x);
59         glVertex2i(xc + x, yc - y);
60         glVertex2i(xc - x, yc - y);
61         glVertex2i(xc - y, yc - x);
62         glVertex2i(xc - y, yc + x);
63         glVertex2i(xc - x, yc + y);
64
65         if (p < 0)
66         {
67             x++;
68             p = p + 2*x + 1;
69         }
70         else
71         {
72             x++;
73             y--;
74             p = p + 2*x - 2*y + 1;
```

```

main.cpp X main.cpp X
75     p = p + 2*x - 2*y + 1;
76     }
77     }
78 }
79
80 void display()
81 {
82     glClear(GL_COLOR_BUFFER_BIT);
83     glPointSize(5);
84
85     glColor3f(1.0, 0.0, 0.0);
86     glBegin(GL_POINTS);
87
88     // Triangle
89     line(200,400,150,150);
90     line(200,300,150,50);
91     line(400,300,150,50);
92
93     // Circle
94     glColor3f(0,1,1);
95     circle(50,300,200);
96
97     glEnd();
98     glFlush();
99 }
100
101 int main(int argc, char *argv[])
102 {
103     glutInit(&argc, argv);
104     glutInitWindowSize(640,480);
105     glutInitWindowPosition(100,100);
106     glutInitDisplayMode(GLUT_RGB | GLUT_SINGLE);
107
108     glutCreateWindow("DDA Line + Circle");
109     glClearColor(1.0,1.0,1.0,1.0);
110
111     glMatrixMode(GL_PROJECTION);
112     gluOrtho2D(0.0,640.0,0.0,480.0);
113
114     glutDisplayFunc(display);
115     glutMainLoop();
116
117     return 0;

```

Output:

